

GIS & Data Analytics Technologist | Spatial Databases | ETL & Enterprise GIS

GIS and Data professional with municipal experience integrating large spatial and non-spatial datasets, developing ETL workflows, and delivering dashboards, reporting, and interactive mapping solutions. Skilled in ArcGIS Enterprise, relational databases, data governance practices, QA/QC, and cross-departmental solution development.

Technical Skills

- **GIS:** ESRI ArcGIS Suite (Pro, Online, Enterprise, Survey123, Field Maps), Spatial Databases, Feature Services, Web Map/Dashboards Development, Open source GIS (QGIS, Leaflet),
- **Automation & Software Development:** Python, Javascript, HTML, CSS, ETL (FME and Python based)
- **Data & Analytics:** Relational & Spatial Databases, SQL, QA/QC, worked with JSON/GeoJSON and XML data, Power BI

Work Experience**Data Analytics/GIS Intern | Water & Wastewater System Design & Performance | Halton Region** [June 2025 - Present]

- Consolidated approximately 400 million rows of historical flow monitoring data (10+ years) collected by multiple contractors, originally stored in monthly CSV files with inconsistent schemas and formats.
- Designed and implemented a Python-based ETL workflow to standardize, validate, and transform disparate flow monitoring datasets into structured site-level tables.
- Normalized varying contractor schemas into a unified relational structure aligned with enterprise data standards.
- Migrated processed datasets into an optimized SQL database (~8GB), improving query performance, integrity, and long-term maintainability.
- Applied QA/QC validation checks to identify schema inconsistencies, missing values, and formatting anomalies.
- Documented end-to-end workflows, transformation logic, and data modelling decisions to support reproducibility, governance compliance, and knowledge transfer.
- Developed an internal interactive web GIS application using Leaflet and GeoJSON to visualize and query operational datasets (flooding history, subsidy programs, drainage surveys), enhancing accessibility and cross-departmental data use.
- Supported Survey123 and Field Maps workflows by integrating contractor-collected household drainage survey data into enterprise GIS databases.
- Developed Power BI dashboards, mapping products, and analytical reports used by engineering, management, and senior leadership to evaluate trends and key performance indicators.

GIS Co-Op | Strategic Business Services | Town of Oakville

[May 2023 - Sept 2023]

- Supported corporate GIS operations by creating, updating, and maintaining spatial datasets and enterprise geodatabases.
- Automated geoprocessing and validation workflows (calculation of lot size) using Python and ArcGIS, improving efficiency and data consistency.
- Developed live, interactive dashboards for leaf collection status in the town using ArcGIS Dashboards.
- Performed QA/QC on GIS data and applications, ensuring adherence to municipal standards and cartographic practices.
- Produced maps, analytical outputs, and documentation for planning, infrastructure, and internal reporting.

Education**Geomatics (Honours) | University of Waterloo**

- **Relevant Coursework:** Spatial Databases · Data Modeling · Geospatial Data Science · Advanced GIS Applications

Projects**Applied Machine Learning & Classification Project (GEOG 471 & GEOG 481) | University of Waterloo**

- Developed supervised classification models and performed accuracy assessment for detecting land cover change caused by 2022 Russian Invasion (GEOG 471) and detecting flood prone areas (GEOG 481).
- Produced clean, analysis-ready datasets (tables and shapefiles) for downstream interpretation.